Yatrafy - GenAI Trip Planning Assistant

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Product RoadMap

	Q1			Q2		
	M1	M2	M3	M4	M5	M6
Deliverables			MVP 1	MVP 2		MVP 3
Product	User & problem definition + target segment		MVP 1 validation, beta feedback & scope freeze	MVP 2 UX definition & iteration		
	Feature scoping & prioritization + user journey				MVP 3 utility scope & conversion prioritization	Scale roadmap finalization
Development	Architecture, environment setup & API contract		Stabilization, QA fixes & performance tuning	Save & resume + UX enhancements (MVP 2)		
	Core build: chat flow, transport, stay, booking				Utility integrations & location-aware content	Production hardening & conversion optimization
UI / UX	Journey mapping + low-fi wireframes		Visual design finalization & MVP 1 sign-off	Feedback-driven UX improvements (MVP 2)		
	Usability testing & interaction refinement				Utility UX + location-aware flows	UX polish, accessibility & E2E validation
Testing / QA	Test strategy & QA environment setup	Functional & integration testing	Regression & performance testing (MVP 1)			
				UX + regression testing (MVP 2)	Utility integration & data validation	Load, reliability & release sign-off
Metrics & Analytics	North-star metric & KPI definition	Event tracking & funnel setup	MVP 1 performance analysis			
				Retention & save-flow measurement	Conversion & monetization metrics	Scale reporting & leadership review
Launch / GTM	Internal alignment & beta planning	Closed beta rollout & iteration	MVP 1 public beta & monitoring			
				MVP 2 announcement & adoption push	MVP 3 pre-launch & partner coordination	MVP 3 public launch & amplification

RICE framework - analysing & prioritizing features

We used the RICE framework to prioritize features objectively by balancing user value, business impact, execution confidence, and development effort. Given the constrained 6-month timeline and iterative MVP approach, RICE helped us avoid intuition-driven decisions and ensured we focused on features that deliver the maximum learning and value early.

MVP	Feature	Reach (1–5)	Impact (1–3)	Confidence %	Effort (1–5)	RICE Score
1	Single-Thread Booking Flow	5	3	100%	3	5
1	Vague Multi-Constraint Trip Queries	4	3	90%	2.5	4.32
1	Tailored Itinerary Generation	5	3	95%	3.5	4.07
1	Real-time Multi-modal Search	5	2.5	85%	3	3.54
1	Dynamic Itinerary Side Panel	4	1.5	90%	1.5	3.6
2	Contextual Quick-Action Buttons	5	1.5	90%	2	3.38
2	Persistent Navigation Tabs	4	1.5	85%	2	2.55
2	Save Without Booking	3	2	85%	2.5	2.04
2	Enhanced Side Panel (Tab-Aware)	3	1.5	80%	2	1.8
3	Safety & Emergency Information Access	2	3	80%	2.5	1.92
3	Dietary-Aware Local Discovery	3	2	75%	3.5	1.29
3	Country-Specific Travel Essentials (SIM / FX)	2	1	70%	2.5	0.56

Minimum Viable Product - 1

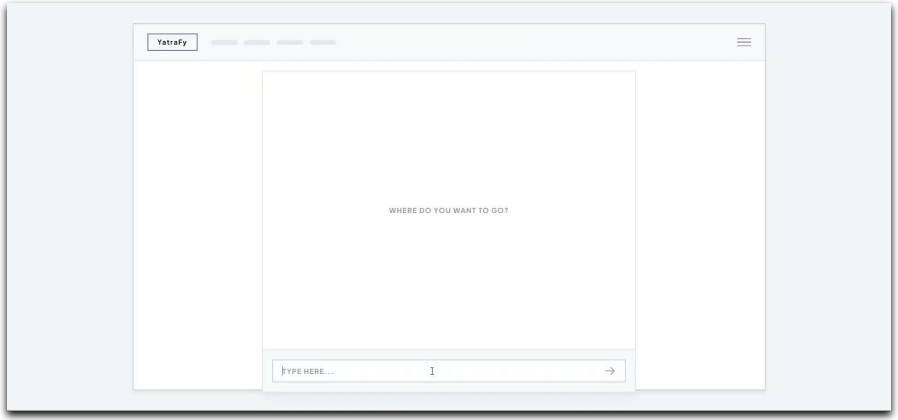
Features:

1. Single-Thread Booking Flow:
   - Enables users to select and book flights/trains and accommodation directly within the chat interface
   - Eliminates the need to navigate across multiple pages during trip planning
2. Vague Multi-Constraint Trip Queries:
   - Allows users to input flexible, open-ended trip requests (e.g., adventure trip for 5 days on a budget)
   - Reduces friction at entry and supports natural planning behavior
3. Tailored Itinerary Generation
   - Generates itineraries based on user preferences collected through conversation
   - Ensures recommendations align with budget, vibe, and activity preferences
4. Real-time Multi-modal Search
   - Compares real-time pricing and availability across Flights, IRCTC Trains, and major Buses
   - Helps users evaluate travel options relevant to the generated itinerary
5. Dynamic Itinerary Side Panel
   - Displays Travel, Stays, and Plan summaries as users make selections
   - Enables quick view, edit, and booking actions without leaving the chat

Success Metrics:

Metric	Def [% of users]	Why
End-to-End Flow Completion Rate	Users completing Travel, Stay & Itinerary steps	Validates core value delivery
AI Trip Summary Acceptance Rate	Users proceeding without major edits	Measures intent understanding
Booking Action Initiation Rate	Users clicking flight or stay booking	Indicates trust & intent

MVP 1 UI:



Minimum Viable Product - 2

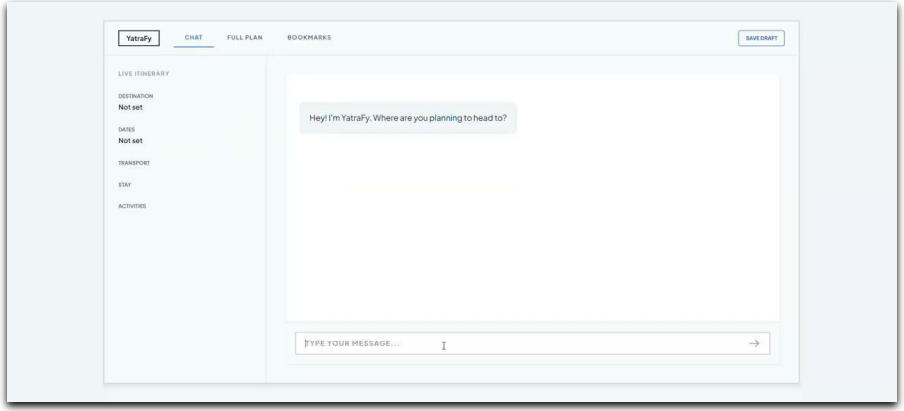
Features:

1. Contextual Quick-Action Buttons:
   - Budget, trip length, number of people, food preference, trip mood
   - Reduces typing and back-and-forth during planning
2. Persistent Navigation Tabs
   - Travel | Stays | Itinerary | Summary
   - Enables non-linear access to earlier decisions without restarting the chat
3. Enhanced Side Panel (Tab-Aware)
   - Displays only relevant information for the active tab
   - Inline edit and book actions without page switches
4. Save Without Booking
   - Allows users to pause and resume planning
   - Reduces pressure at decision points and supports longer consideration cycles

Success Metrics:

Metric	Def [% of users]	Why
Itinerary Completion Rate	Users who reach and finalize the detailed itinerary step	Measures whether UX improvements help users complete planning end-to-end
Planning Experience CSAT	Users rating the planning experience positively after completing a session	Validates improved usability, clarity, and overall planning confidence
Pre-Accommodation Drop-off Rate	Users who exit before reaching or selecting accommodation	Indicates whether reduced friction and save options prevent early abandonment

MVP 2 UI:



Minimum Viable Product - 3

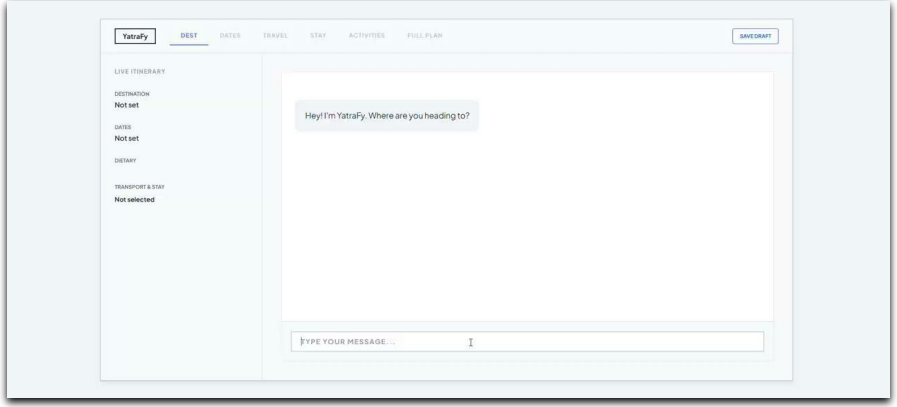
Features:

1. Safety & Emergency Information Access:
   - Location-aware access to nearby hospitals, pharmacies, and police stations directly within the chat
   - Provides verified, essential information to support users in unfamiliar destinations
2. Dietary-Aware Local Discovery
   - Restaurant discovery filtered by dietary preferences (vegetarian, vegan, halal)
   - Reduces food-related anxiety and improves confidence during travel
3. Country-Specific Travel Essentials
   - Instant guidance on currency exchange and local SIM / eSIM options
   - Covers basic cultural and logistical information for first-time or international travelers

Success Metrics:

Metric	Def [% of users]	Why
Post-booking engagement rate	Users interacting with the app after the booking is confirmed	Measures continued value and retention
Repeat usage across trips	Users returning to use the tool for a subsequent trip	Indicates long-term product-market fit
NPS uplift vs non-GenAI users	Net Promoter Score difference between AI and manual users	Quantifies the satisfaction boost from GenAI

MVP 3 UI:



Risk Assessment

Risks are identified proactively at each MVP stage using structured AI, journey, and security risk frameworks, combined with real-time product metrics and user feedback.

Risk Identification

Framework	Risk Lens	How It Identifies Risk in Yatrafy
NIST AI RMF	AI Trustworthiness	Mitigating hallucinations in IRCTC/Flight data and model bias.
FMEA	Journey Reliability	Identifying failure points in "Single Thread Booking" and crisis alerts.
STRIDE	Data Security	Protecting sensitive consumer data.

Contingency Plan

Risk Area	Threat examples	Contingency Plan
Accuracy *(MVP 1–2)*	Incorrect IRCTC/Flight data.	**Real-Time Sync:** Mandate live API validation for all bookings.
Reliability *(MVP 1–3)*	Unreliable last-mile transport or scams.	**Verified Concierge:** Provide a pre-vetted local transport database and instant 24/7 human escalation.
Security *(All MVps)*	Exposure of PII or IRCTC credentials.	**Local LLM Training:** Use proprietary models (Llama 3/Mistral) to keep sensitive data off public servers.
Continuity *(MVP 2–3)*	Sudden cancellations of flights/hotels.	**Proactive Crisis Management:** Automated system for instant re-booking and compensation filing.
Adoption *(MVP 2)*	Low engagement from non-English users.	**Localized Fallback:** Deploy Indic LLMs (Krutrim/Sarvam-1) for planning in major Indian languages.

Thank you

Varsha Sivaramakrishnan

Lekha Chenguttuvan

Gunanicaa Arun

Team Code Pink

AI Usage: Google Gemini was used to create the UI for all the MVPs, and to understand process that goes behind creating a product.